Research Report – First Power Forecasting Prototype

1. Introduction

The goal of this first prototype was to investigate whether a machine-learning model can predict power production more accurately than a simple Copy-Yesterday baseline.

The prototype was developed using Python and scikit-learn. It works directly with the historical CSV datasets provided for the project.

For this prototype, no Ollama, LLM, cloud API, API key, Raspberry Pi or external hosting was used. The purpose was first to determine whether a traditional machine-learning approach could provide useful forecasting results before researching more complex deployment options.

2. Dataset

The prototype inspected the following datasets:

- `buildings.csv`
- `electricity.csv`
- `weather.csv`

The selected timestamp column was:

`timestamp_utc`

The selected prediction target was:

`production_w`

`production_w` was selected because it represents the measured power production/output in watts.

The measurements in the dataset are mostly recorded at 15-minute intervals.

3. Data Preparation

Because this is a time-series forecasting problem, the data was sorted chronologically before training and testing.

The dataset was divided using an 80/20 chronological split:

- First 80% of timestamps → training data
- Last 20% of timestamps → test data

The data was deliberately not randomly shuffled. Randomly mixing older and newer timestamps would not represent the way the model would be used for forecasting future values.

4. Copy-Yesterday Baseline

The Copy-Yesterday method was used as the baseline.

For each timestamp, this method predicts power production using the actual production value from the same building at the same time one day earlier.

The implementation matches:

```
timestamp_utc - 1 day
```

with the corresponding historical timestamp.

This is more reliable than simply shifting the dataset by a fixed number of rows, because it uses the timestamps themselves.

The baseline provides a simple reference point. A more advanced forecasting model should ideally perform better than this method to show that the additional complexity provides value.

5. Random Forest Model

The machine-learning model selected for this first prototype was the RandomForestRegressor from scikit-learn.

The configuration used was:

Setting	Value
Model	RandomForestRegressor
Number of trees	200
Random state	42
Train/test method	Chronological 80/20 split
Prediction target	`production_w`

The Random Forest used available information including timestamp features, historical weather data, building information and previous production values to learn patterns related to power production.

6. Evaluation Method

The Random Forest and Copy-Yesterday baseline were evaluated on the same test timestamps to make the comparison fair.

Two main evaluation metrics were used.

MAE (Mean Absolute Error) measures the average size of the prediction error. A lower MAE means that predictions are, on average, closer to the actual values.

RMSE (Root Mean Squared Error) also measures prediction error but gives more weight to larger errors. A lower RMSE therefore indicates fewer or smaller large prediction errors.

MAPE (Mean Absolute Percentage Error) was also considered as an evaluation metric. However, MAPE needs to be used carefully when actual values are zero or close to zero. For this first comparison, the reported results focus on MAE and RMSE.

7. Results

The prototype produced the following results:

Forecasting Method	MAE	RMSE
Copy-Yesterday	10,746.70 W	22,209.85 W
Random Forest	4,673.03 W	10,071.16 W

The Random Forest performed better than Copy-Yesterday on both evaluation metrics.

The average absolute prediction error decreased from approximately 10.7 kW to 4.7 kW.

Based on these reported values, this is a reduction in MAE of approximately 56.5% compared with the baseline. RMSE decreased by approximately 54.7%.

This indicates that, on this particular test split, the Random Forest was able to produce predictions that were closer to the actual historical values than the Copy-Yesterday method.

8. Interpretation

The results of the first prototype are promising.

The Random Forest appears to have learned useful patterns from the available historical information, including timestamp features, weather data, building information and previous production values.

The lower MAE means that the model's predictions had a smaller average absolute error than the Copy-Yesterday predictions.

The lower RMSE is also important because RMSE penalizes larger prediction errors more strongly. The reduction therefore shows that the Random Forest also reduced the effect of larger errors compared with the baseline.

For this prototype, the results support continuing research into traditional machine-learning forecasting methods rather than using an LLM or Ollama for numerical forecasting itself. Ollama and LLM-based tools were not tested in this prototype, so this experiment does not provide a direct performance comparison between Random Forest and an LLM.

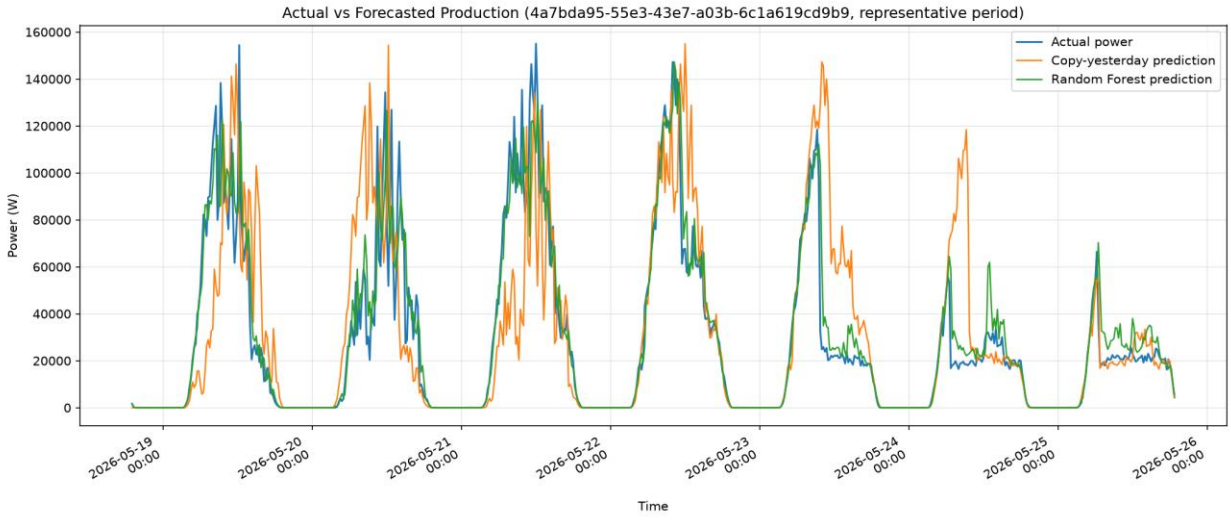
The results also do not prove that Random Forest is the best forecasting model. They only show that the tested Random Forest implementation performed better than the Copy-Yesterday baseline in this first experiment.

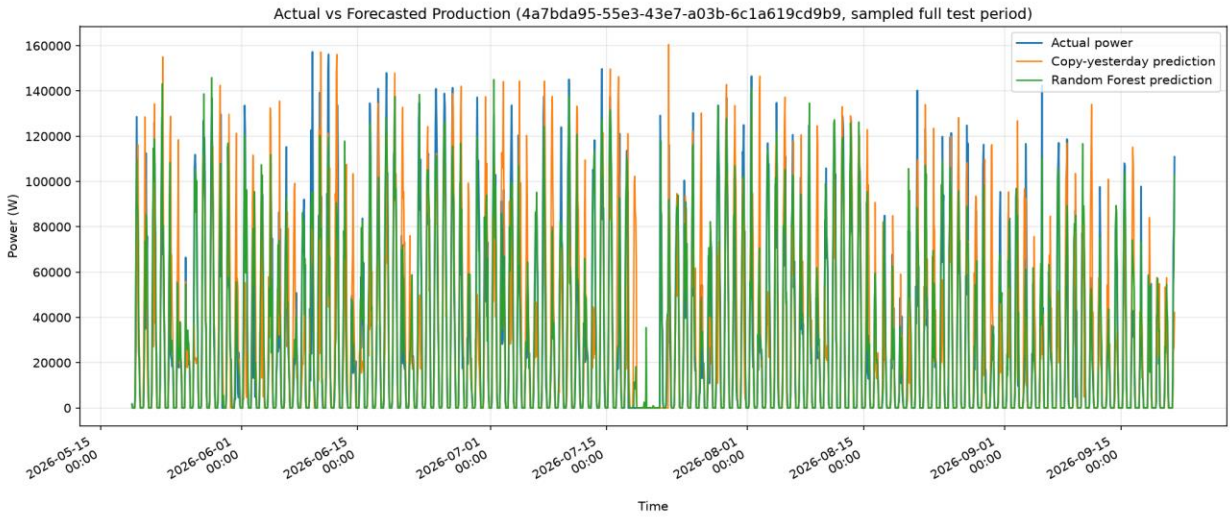
9. Prototype Outputs

The prototype generates several files that can be used to inspect the results:

Output	Purpose
`output/predictions.csv`	Contains the generated prediction results
`output/forecast_chart.png`	Visualisation of the forecast
`output/forecast_chart_full_sampled.png`	Sampled visualisation of the larger forecast result
`models/random_forest_model.joblib`	Saved trained Random Forest model

The original CSV datasets were not modified by the prototype.





10. Conclusion

The first prototype successfully demonstrated that a local Python/scikit-learn machine-learning approach can be used for power forecasting with the provided historical datasets.

On the tested historical period, Random Forest clearly outperformed the Copy-Yesterday baseline:

Copy-Yesterday:

MAE: 10,746.70 W

RMSE: 22,209.85 W

Random Forest:

MAE: 4,673.03 W

RMSE: 10,071.16 W

This prototype should be seen as a first experiment to test whether our forecasting approach works and whether a machine-learning model can improve on the Copy-Yesterday baseline. For this experiment, we focused on predicting power production (`production_w`).

The results showed that the approach is promising, but predicting production is not the main goal of the project. The next step is to apply what we learned from this experiment to

energy usage/consumption forecasting. The next prototype will therefore focus on predicting how much electricity a building is expected to use.